

Homework Assignment 1

Solution Manual

Solutions to all eight questions. Equivalent correct arguments and circuits are acceptable.

Question 1 Building with NAND gates

Each gate equation below represents one physical two-input NAND gate. An intermediate output may feed more than one input or gate.

(a) **NOT uses 1 gate.**

$$Y = \text{NAND}(A, A) = \neg(A \wedge A) = \neg A.$$

(b) **AND uses 2 gates.**

$$t = \text{NAND}(A, B), \quad Y = \text{NAND}(t, t).$$

Then $Y = \neg t = \neg \neg(A \wedge B) = A \wedge B$.

(c) **OR uses 3 gates.**

$$u = \text{NAND}(A, A), \quad v = \text{NAND}(B, B), \quad Y = \text{NAND}(u, v).$$

Since $u = \neg A$ and $v = \neg B$, De Morgan's law gives $Y = \neg(\neg A \wedge \neg B) = A \vee B$.

Question 1 continued**(d) XOR uses 4 gates.**

$$\begin{aligned}
 t &= \text{NAND}(A, B), & u &= \text{NAND}(A, t), \\
 v &= \text{NAND}(B, t), & Y &= \text{NAND}(u, v).
 \end{aligned}$$

The output t is shared by the gates producing u and v . Using De Morgan's law and distributivity, we obtain

$$\begin{aligned}
 Y &= \neg(u \wedge v) \\
 &= \neg u \vee \neg v \\
 &= (A \wedge t) \vee (B \wedge t) \\
 &= (A \vee B) \wedge \neg(A \wedge B) \\
 &= (A \vee B) \wedge (\neg A \vee \neg B) \\
 &= (A \wedge \neg B) \vee (\neg A \wedge B).
 \end{aligned}$$

The last step also uses the complement and identity laws to remove the terms $A \wedge \neg A$ and $B \wedge \neg B$. The resulting expression is exactly XOR: one input is true and the other is false.

Evaluating the gate equations for all four constructions gives the following truth table. Each output column matches the corresponding operation:

A	B	Y_{NOT}	Y_{AND}	Y_{OR}	Y_{XOR}
0	0	1	0	0	0
0	1	1	0	1	1
1	0	0	0	1	1
1	1	0	1	1	0

Here NOT acts on A ; its output is independent of B . The gate counts are **1, 2, 3, 4** for NOT, AND, OR, and XOR, respectively.

Question 2 Proving an arithmetic equality

We use $0 + m = m$ and $S(n) + m = S(n + m)$, which recurse on the *first* input of addition. First,

$$\begin{aligned}
 1 + 2 &= S(0) + 2 && \text{definition of 1} \\
 &= S(0 + 2) && \text{successor rule} \\
 &= S(2) && \text{starting rule} \\
 &= 3 && \text{definition of 3.}
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 2 + 1 &= S(1) + 1 && \text{definition of 2} \\
 &= S(1 + 1) && \text{successor rule} \\
 &= S(S(0) + 1) && \text{definition of 1} \\
 &= S(S(0 + 1)) && \text{successor rule} \\
 &= S(S(1)) && \text{starting rule} \\
 &= S(2) && \text{definition of 2} \\
 &= 3 && \text{definition of 3.}
 \end{aligned}$$

Both expressions equal 3, so symmetry and transitivity of equality give $1 + 2 = 2 + 1$. No commutativity law was used.

Question 3 Direct proofs

(a) Let a and b be arbitrary odd integers. By definition, $a = 2u + 1$ and $b = 2v + 1$ for some $u, v \in \mathbb{Z}$. Therefore,

$$\begin{aligned}
 a^2 + b^2 &= (2u + 1)^2 + (2v + 1)^2 \\
 &= 4u^2 + 4u + 4v^2 + 4v + 2 \\
 &= 2(2u^2 + 2u + 2v^2 + 2v + 1).
 \end{aligned}$$

The expression in parentheses is an integer, so $a^2 + b^2$ is even.

(b) Let $a, b, c, r, s \in \mathbb{Z}$, and assume $a \mid b$ and $a \mid c$. There are integers u, v such that $b = au$ and $c = av$. Thus

$$rb + sc = r(au) + s(av) = a(ru + sv).$$

Since $ru + sv \in \mathbb{Z}$, the definition of divisibility gives $a \mid (rb + sc)$. This proof also covers $a = 0$; it never divides by a .

Question 4 Proof by contrapositive

(a) The original conclusion is a conjunction. By De Morgan's law,

$$\neg(m \text{ is odd} \wedge n \text{ is odd}) \equiv (m \text{ is not odd}) \vee (n \text{ is not odd}).$$

Using the even-odd dichotomy, the contrapositive is:

For all integers m, n , if m is even or n is even, then mn is even.

Here “or” includes the possibility that both are even.

(b) Let $m, n \in \mathbb{Z}$ and assume that m is even or n is even. If m is even, write $m = 2k$ with $k \in \mathbb{Z}$. Then

$$mn = 2(kn),$$

which is even because $kn \in \mathbb{Z}$. If n is even, write $n = 2\ell$ with $\ell \in \mathbb{Z}$. Then

$$mn = 2(m\ell),$$

which is even because $m\ell \in \mathbb{Z}$. These cases cover every possibility in the hypothesis, including the case in which both are even. Thus the contrapositive holds, and so does the original implication.

Question 5 Proof by contradiction

(a) Suppose, for a contradiction, that there is a greatest odd integer M . Since M is odd, $M = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$M + 2 = 2(k + 1) + 1.$$

Since $k + 1 \in \mathbb{Z}$, the integer $M + 2$ is odd. But $M + 2 > M$, contradicting the assumption that M is the greatest odd integer. Therefore there is no greatest odd integer.

(b) Suppose, for a contradiction, that integers d, n satisfy $d > 1$, $d \mid n$, and $d \mid (n + 1)$. By the definition of divisibility, there are integers a, b such that $n = da$ and $n + 1 = db$. Subtracting gives

$$1 = (n + 1) - n = db - da = d(b - a).$$

Because $d > 0$ and $d(b - a) = 1 > 0$, the integer $b - a$ is positive, so $b - a \geq 1$. Consequently,

$$1 = d(b - a) \geq d > 1,$$

which is a contradiction. Thus no integer greater than 1 divides two consecutive integers.

Question 6 Proof by cases

Let $n \in \mathbb{Z}$ be arbitrary. By the fact given in the question, n has exactly one of the forms $3k$, $3k + 1$, or $3k + 2$ for some $k \in \mathbb{Z}$. We consider all three cases.

Case 1: $n = 3k$. Then

$$n(n+1)(n+2) = 3[k(n+1)(n+2)].$$

The expression in brackets is an integer, so the product is divisible by 3.

Case 2: $n = 3k + 1$. In this case $n + 2 = 3(k + 1)$, so

$$n(n+1)(n+2) = 3[n(n+1)(k+1)].$$

Again, the expression in brackets is an integer, so the product is divisible by 3.

Case 3: $n = 3k + 2$. Now $n + 1 = 3(k + 1)$, and hence

$$n(n+1)(n+2) = 3[n(k+1)(n+2)].$$

The expression in brackets is an integer, so the product is divisible by 3 in this case as well.

These cases exhaust all integers, including negative integers and zero. In every case, at least one of the three consecutive factors is a multiple of 3. Therefore $3 \mid n(n+1)(n+2)$ for every integer n .

Question 7 Nested quantifiers

(a) Statement A is true. It says that every integer has an integer that sums with it to zero. Let $x \in \mathbb{Z}$ be arbitrary and choose $y = -x$. Since $-x \in \mathbb{Z}$ and $x + (-x) = 0$, the required witness exists for every x .

Statement B is false. It says that a single integer sums to zero with every integer. If such a y existed, the choice $x = 0$ would give $y = 0$. But the choice $x = 1$ would then give $1 + 0 = 0$, a contradiction. Thus no one value of y works for every x .

Statement C is false. It says that every integer is twice an integer. Take $x = 1$. For any integer y , $2y$ is even, whereas 1 is odd. Therefore no integer y satisfies $2y = 1$, giving a counterexample to the universal statement.

Statement D is true. It says that there is an integer whose product with any integer equals that second integer. Choose $x = 1$. For every $y \in \mathbb{Z}$, the multiplicative identity law gives $1 \cdot y = y$. This one witness works for all y .

(b) Negation of A.

$$\exists x \in \mathbb{Z} \forall y \in \mathbb{Z} (x + y \neq 0).$$

This says some integer has no additive inverse among the integers.

Negation of B.

$$\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} (x + y \neq 0).$$

This says that every proposed y fails for some integer x . For example, after y is given, the choice $x = 1 - y$ makes $x + y = 1 \neq 0$.

(c) In A, x is chosen first, so the witness y may depend on that x ; $y = -x$ is allowed. In B, one y must be chosen before the universal choice of x , and that same y must work for every x . The order of the quantifiers changes the requirement and, here, the truth value.

Question 8 Translating English into logic

All variables in this question range over the domain D of people. Recall that $S(x)$ means x is a student in the course, $T(x)$ means x submitted the assignment, and $H(x, y)$ means x helped y .

(a) Every student submitted the assignment.

$$\forall x [S(x) \Rightarrow T(x)].$$

Its negation is

$$\exists x [S(x) \wedge \neg T(x)].$$

In words, some student did not submit the assignment. The implication in the original statement restricts the submission requirement to students.

(b) Some student helped every other student.

$$\exists x [S(x) \wedge \forall y [(S(y) \wedge y \neq x) \Rightarrow H(x, y)]].$$

Negating the existential statement, the conjunction, and then the universal statement gives

$$\forall x [\neg S(x) \vee \exists y [S(y) \wedge y \neq x \wedge \neg H(x, y)]].$$

This says that every student has some other student whom they did not help. The $\neg S(x)$ alternative ensures that the formula makes no demand of a person who is not a student.

(c) Every student was helped by a different student.

$$\forall x [S(x) \Rightarrow \exists y [S(y) \wedge y \neq x \wedge H(y, x)]].$$

Its negation is

$$\exists x [S(x) \wedge \forall y [\neg S(y) \vee y = x \vee \neg H(y, x)]].$$

In words, there is a student who was not helped by any other student. The predicate is $H(y, x)$ because y is the helper and x receives the help.

Negation rules used. Switch $\forall$ and $\exists$ as a negation moves inward, use De Morgan's laws, and replace $\neg(A \Rightarrow B)$ with $A \wedge \neg B$. Negating $y \neq x$ gives $y = x$. The displayed negations contain no negated compound formulas.

Vacuous cases. If there are no students, the original statements in (a) and (c) are true and (b) is false. If there is exactly one student, (b) is true because there are no other students to help, while (c) is false because its helper must be a different student. The formulas preserve these distinctions.